



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

Tactile Time: Interactive 3D Heritage and Art Experiences with TouchDesigner

Eslam ElShamy¹, and Mohamed W. Fareed¹

¹ Co-founder, LO(C)AI – Lab of Creative/Critical AI, Germany

Keywords: Extended Reality (XR), FAIR Data Principles, Digital Paradata, Semantic Modelling, Participatory Pedagogy

Abstract

“Tactile Time” investigates how interactive 3D visualisation and XR technologies can transform education, public engagement, and the interpretation of cultural heritage. Using TouchDesigner as a technical platform, the project converts historical datasets—specifically test 3D datasets of Islamic architectural patterns from Cairo—into immersive experiences. It allows users to explore and manipulate spatial elements in real time, reflecting current advancements in web-based and interactive 3D engagement (Web3D, 2024).

The project addresses the research question: How can interactive 3D/XR environments facilitate collaborative learning while adhering to FAIR principles and maintaining cultural heritage data best practices (Amico & Felicetti, 2024)? Participants navigate virtual reconstructions, where interactive elements directly contribute to educational outcomes. Rather than viewing decorative, static displays, users manipulate structural parameters to uncover hidden relationships. XR enhances spatial immersion, while AI systems act as active collaborators in real-time pattern generation and knowledge production, turning passive observation into active spatial problem-solving (Champion, 2015).

Methodologically, the project integrates these 3D datasets with semantic modelling approaches, utilising the CIDOC Conceptual Reference Model (CRM) ontology to ensure robust data structures. TouchDesigner’s node-based workflow enables modular manipulation while preserving transparency. To firmly engage with all FAIR principles and align with wider 3D accessibility initiatives (DFG 3D-Viewer, 2024), the project stores datasets in open-access repositories (e.g., Zenodo) to maximise findability. Datasets and reconstructions are rigorously documented using Dublin Core metadata standards and

structured paradata frameworks aligned with the London Charter (Koutsoudis, Arnaoutoglou, & Pavlidis, 2018). This ensures a clear, transparent association between the 3D models and their source data, guaranteeing reproducibility, interoperability, and long-term preservation. The proposed interactive workflow is summarised in Figure 1.

Preliminary outcomes, measured through qualitative and quantitative participant feedback during physical co-creation workshops, indicate clear educational improvements. Participants demonstrated enhanced spatial retention and a deeper structural understanding of heritage spaces. Furthermore, they engaged critically with the ethics and responsibilities of working with 3D data. Ultimately, “Tactile Time” demonstrates how interactive 3D and XR experiences can effectively bridge archival research, pedagogy, and public participation, fostering sustainable and responsible engagement with heritage.

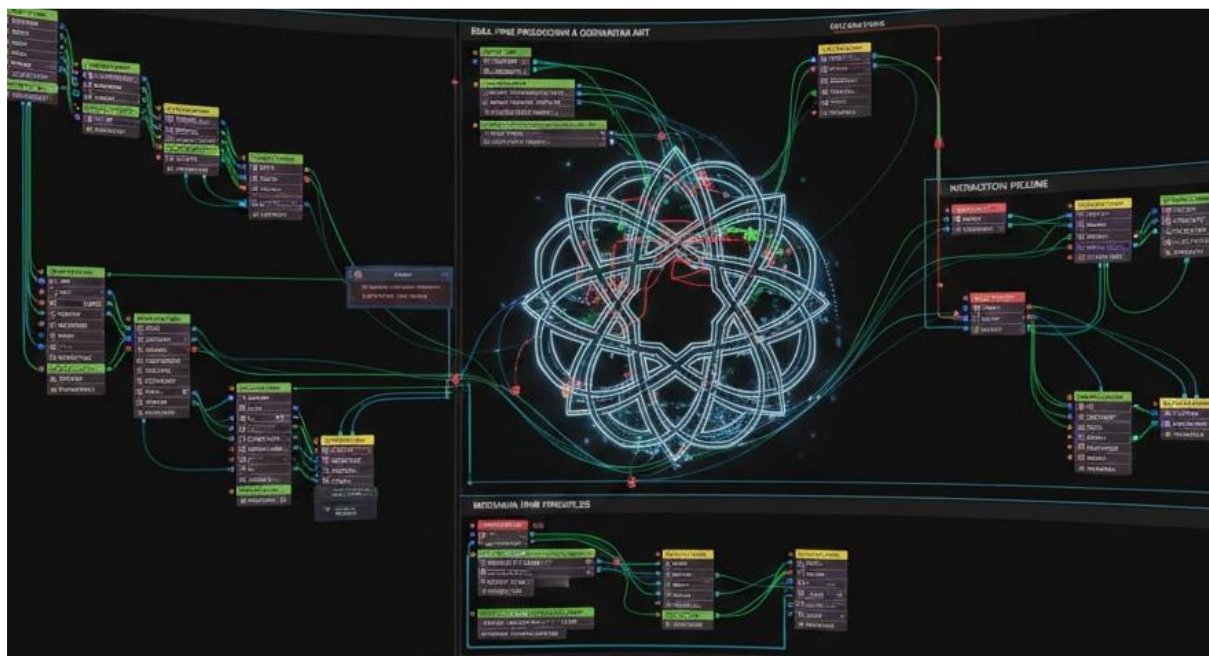


Fig. 1 A 3D reconstruction of Islamic pattern in TouchDesigner, highlighting interactive exploration of structural and decorative features.

References

- Amico, N., & Felicetti, A. (2024). 3D data long-term preservation in cultural heritage. eArchiving Initiative, European Commission. Retrieved from <https://digital-strategy.ec.europa.eu/en/library/report-long-term-preservation-3d-data-cultural-heritage>
- Champion, E. (2015). Critical gaming: Interactive history and virtual heritage. Routledge.
- DFG 3D-Viewer. (2024). Second funding phase. Retrieved from <https://dfg-viewer.de/en/second-funding-phase>
- Koutsoudis, A., Arnaoutoglou, F., & Pavlidis, G. (2018). 3D cultural heritage documentation: Methods and applications. *Journal of Cultural Heritage*, 33, 1–12.
- Web3D. (2024). Web3D conference report. Retrieved from <https://www.web3d.org/news-story/web3d-2024-conference-report>